

In Deep Water: Designing Contrasting Temporal Modes for Collaborative, Embodied Engagement with Marine Plastic Pollution

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Figure 1: Two players use our custom physical controller to collaboratively explore *In Deep Water*, a digital marine biorama that invites novel, tactile engagement with fragile aquatic ecosystems

Abstract

Marine plastic and microplastic pollution can be difficult to communicate because its impacts unfold slowly, are geographically dispersed, and often remain imperceptible. We address this challenge through *In Deep Water*, a public-facing interactive installation that translates marine science research into collaborative exploration. The system combines a sculpted virtual marine biorama with custom-built physical controllers that allow players to navigate simulated waterways, locate discarded research beacons, and decode situated data from studies conducted in Aotearoa, New Zealand. The installation is developed in two complementary modes, one designed for open-ended exploration and the other structured around time-limited play, to investigate how different interaction conditions shape engagement and sense-making. Our contribution is a replicable

design approach for turning environmental datasets into embodied, social experiences, supporting HCI researchers and practitioners designing data-driven public installations and educators and cultural institutions seeking accessible formats for research translation and environmental engagement.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; *Empirical studies in HCI*; *Haptic devices*; *User studies*; **Empirical studies in interaction design**.

Keywords

Interactive installation; embodied interaction; research-through-design; sustainability; environmental games; collaborative play; slowness; replayability

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1 Introduction

Plastic pollution presents a persistent challenge to public participation. Its causes are distributed across systems of production and consumption, its impacts are often invisible, and its consequences unfold gradually across ecosystems and generations. Although scientific research on plastic and micro-plastic pollution has expanded rapidly, much of this knowledge remains inaccessible to non-specialist audiences and is typically communicated through academic publications, policy reports, or abstract data visualisation.

In Deep Water [13] responds to this challenge by treating games and interactive installations as legitimate modes of research translation within human-computer interaction (HCI). The project emerges from a five-year MBIE-funded broader research program investigating how the plastics ecosystem in Aotearoa, New Zealand, might be reshaped toward greater circularity. Although much of that program is focused on system modelling and policy-orientated tools, the design team identified a parallel need to engage the broader public with the environmental stakes of plastic waste through accessible, affective, and interactive media.

We focus on marine ecosystems because they combine perceived pristine beauty with systemic degradation and because microplastics are both pervasive and largely imperceptible. Communicating this form of harm requires approaches that move beyond deficit-based information delivery and avoid overwhelming audiences with scale, inevitability, or moral prescription. Rather than positioning players as agents who optimise or “solve” environmental problems, *In Deep Water* invites participants to collaboratively explore, observe, and interpret environmental change as an unfolding condition through embodied interaction, shared discovery, and aesthetic immersion.

The demo is designed as a research instrument rather than as a static exhibit. It presents two installations that run the same simulated world, but differ in temporal framing: an open-ended mode that supports slow exploration and a timed mode that encourages short runs and replays. This paired configuration enables comparative observation of how pacing and physical affordances shape co-located interaction strategies, division of labour, and interpretive sense-making around environmental content. In this paper, we present *In Deep Water* as an interactive prototype ready for play-testing with an expert HCI community, describe the system design and lineage, and outline how comparative demonstration supports research-through-design in public exhibition contexts.

2 Related Work

In Deep Water builds on previous work in games, embodied environmental installations, and Sustainable HCI.

Commercial and indie games such as *Tamagotchi* [1] and *Cloud Gardens* [11] highlight how slow, minimal interaction can sustain attention and communicate gradual system change without optimising for challenge. This inspired our use of gentle temporal dynamics and interpretive play, along with interface traditions in *Dr. Umgebung's School of Life* [3] and *In Other Waters* [6] that frame interaction as reading a system rather than mastering it.

In research installations, Sustainable HCI emphasises situated engagement beyond solutionist behaviour-change approaches, with Perovich et al. noting that oceans remain under-represented in the field [12]. Related systems show how embodied interfaces

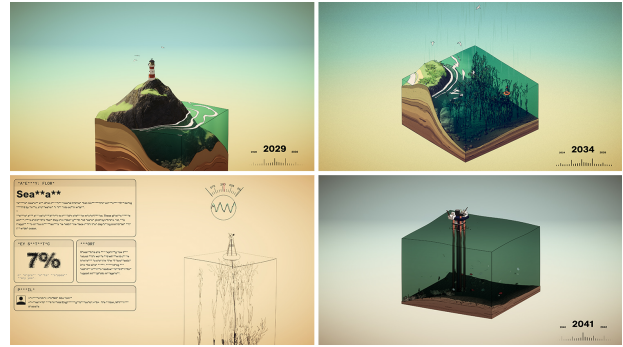


Figure 2: Screenshots illustrating different views of the simulated environments and beacon locations over time and an example of a decoding interface.

can shape attention and affect, including underwater embodiment in VR [10] and tangible metaphors that render ecological processes perceptible through bodily coupling [8, 14]. These examples informed our focus on embodied interaction while motivating a design that preserves interpretive distance rather than relying on seamless immersion.

Finally, several projects explore how environmental data is made legible to the public. Ferreira et al. demonstrate that data humanism approaches such as personalisation and positive narrative framing can increase engagement and help relate large-scale climate phenomena to individual experience [4, 5]. *In Deep Water* takes a complementary approach by foregrounding tactile friction and co-located sense-making, asking visitors to actively tune into marine data through the effort of interaction rather than through automated explanation.

3 System Overview

In Deep Water combines a virtual marine simulation with a custom-built physical controller to support co-located, interpretive interaction.

3.1 Virtual Marine Biorama

The virtual component of *In Deep Water* is an explorable marine **biorama** - a diorama re-imagined as a playable simulation of a living ecosystem. It draws on the waterways of Aotearoa, New Zealand, spanning estuaries, coastlines, kelp forests, mangroves, coral reefs, and deep ocean environments. The world uses an isometric sectioned perspective that lets players observe surface and subsurface dynamics simultaneously. Over time, environmental conditions gradually degrade: plastic accumulates, species diminish, and visual and audio cues signal increasing ecological stress. These changes are intentionally incremental, encouraging players to notice slow transformation rather than dramatic collapse. Screenshots of the biorama are illustrated in Figure 2.

The abstract style of the digital environment, smooth transitions between locations, and the polish of visual and audio assets is deliberately designed to be in stark contrast to the seemingly hacked design of the physical box (see Figure 3) which is constructed from waste materials and modified controllers. This contrast is used as a narrative device to indicate an obvious difference and disconnect between those responsible for the marine environment and the maker of the curious box. This way the players get a sense that they are intervening in the virtual



Figure 3: Latest Controller Box V3.0.

world as a subversive co-conspirator associating the otherwise simple simulation to cinematic tropes familiar in science-fiction.

3.2 Physical Control System

A core feature of *In Deep Water* is its custom-built physical controller box equipped with an Arduino and several modified dials, sliders, levers, buttons, and LED indicators (Figure 4). Using a combination of zoom, tilt, rotate, and directional controllers, players can navigate through various marine environments and follow living creatures such as whales, birds, fish, and turtles. There is a button to scan for hidden beacons with lights to help guide players to their precise location, and a turn dial to decode the lost beacon data. The box also has a series of lights that illuminate when a beacon is successfully found and indicate how many more are left. Providing both navigation controls and progress feedback on the box replaces the need for complex graphical interface components and squarely anchors the interaction and achievements with the player in the physical world rather than being dislocated from the player in the virtual one.

The controller box is large enough for at least two users to coordinate play, and multiple headsets are attached to further support this. Co-present players are encouraged to verbalise what they notice, propose interpretations, and coordinate actions at the controller, turning interaction into shared sense-making.

The design deliberately resists conventional game controllers and instead draws on the visual language of scientific instruments, improvised machinery, and hacker-style fabrication (Figure 3). This framing positions the box as both unfamiliar and recognisable, inviting curiosity and hands-on experimentation. Beyond its aesthetic role, the controller acts as an external cognitive artefact that distributes interpretation between hands, bodies, and conversation. By requiring coordinated manipulation, it encourages negotiation, explanation, and shared hypothesis-making between co-present players.

3.3 Decoding as Embodied Interaction

When a player approaches a beacon, it appears as a scrambled signal rather than readable text or a clear icon. To reveal its content, players must use the physical controller to *tune* the beacon by trying different combinations of dials and sliders until the display resolves to a legible state. In other words, the player’s task is to iteratively adjust the controls, observe the system’s feedback, and collaboratively work out how to “lock in” the correct configuration.

This interaction treats information access as a hands-on interpretive process rather than a button press. It slows the pace of discovery, encourages experimentation and discussion between

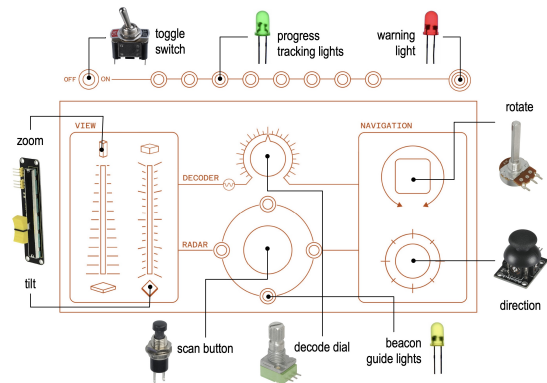


Figure 4: Controller interaction diagram showing original state of bought controllers before being modified in the fabrication lab

co-present players, and reinforces the framing of the visitor as an investigator who must separate signal from noise. Once decoded, each located beacon reveals (1) a short warning about the impacts of plastic pollution on that site (the crisis), (2) a brief profile of a researcher and their efforts to mitigate the harmful impacts (the hope), and (3) short but significant statistics that are easy for the players to later recall and retell. The pairing of harm with ongoing scientific work connects the data to real-world expertise and positions the encounter as both a moment of concern and a prompt for attentive engagement and continued conversations.

4 Early Public Play-testing

In Deep Water follows a research-through-design approach [17], treating public deployment as a generative part of system development rather than a final validation step. From early prototypes onwards, the installation has been designed for museum contexts, where co-located play and informal talk provide a natural setting for observing how visitors interpret the system.

We play-tested a high-fidelity prototype of an earlier version at the Australian Centre for the Moving Image (ACMI) Audience Lab in June 2025 and March 2026, with more than 1,000 visitors engaging with the installation each day. Data collection combined brief surveys, informal interviews, and structured observation by the design team. This aligns with established HCI practices that treat exhibitions and interactive installations as sites of inquiry and iterative refinement rather than controlled evaluation [9, 16].

In all play-tests, visitors responded positively to tactile controls and cooperative mode of interaction, often using the controller as a shared reference point for discussion and joint experimentation. At the same time, observations surfaced recurring challenges around feedback legibility, beacon discoverability, and interpreting the function of specific controls. These findings directly informed subsequent design iterations, including a dual-installation structure that supports comparative reflection on how different temporal framings shape attention and sense-making within the same underlying system. The next phase of refinement is focused on play-testing with HCI experts to further assess legibility and interpretive accessibility.

5 Latest Iteration, User Experience and Interaction Flow

For CHI 2026, we present *In Deep Water* as a side-by-side interactive demo with two parallel installations. Each installation includes a display and a custom controller box and supports cooperative two-player use (with optional single-player play). Both instances run the same underlying simulation and content, but differ in temporal framing to enable direct comparison. The first installation offers an open-ended mode with no explicit time limits, which supports slow exploration, observation, and discussion. The second presents the same world through short, time-limited runs that encourage replay, comparison, and light competition. By keeping the environment and mechanics constant, the demo isolates the pace as the primary variable and allows attendees to experience how temporal constraints shape attention, interaction strategies, and interpretive talk.

5.1 Intended Experience and Interaction Flow

Visitors approach the first installation as an open, non-prescriptive system. The game begins without a tutorial; instead, participants learn through the controller features and feedback (see Section 3.2), and the gradual reveal of the system responses. The players use the controller to navigate the environments, locate beacons, and contemplate decoded messages. This version emphasises exploration and discovery over task completion, while still providing concrete interaction goals through beacon finding and decoding. After completing a session, participants can optionally complete a brief feedback card to capture perceived clarity, affective tone, and moments of surprise or insight.

The same players are then invited to try the second installation, enabling an immediate comparison between open-ended and timed engagement with the same content. This version is similar to the first in most ways, except for an additional timing component and warning system. Play begins as soon as any user engages with a box controller, and immediately a time bar is triggered, indicating a rapid increase in passing years along with a health bar showing a rapid decline in the ecosystem health. Simultaneously, plastic pollution is piling up in the pristine marine environments, the number of living creatures is diminishing, plant-life is in decline, and a red warning beacon is flashing both in the virtual world and on the controller box (red LED). Players rarely complete all beacons within the allotted time. At the end of a session, players can replay the game to improve their own performance or surpass that of another player, introducing a light competitive dynamic in this version.

5.2 Observational Focus and Design Learning Goals

The comparison between the two installation experiences is intended to surface whether visitors prefer unhurried interpretive play or time-bounded urgency, and how each framing influences what people attend to, discuss, and remember. Across both installations, we attend to interaction breakdowns (e.g., misunderstood controls, unclear feedback, missed beacons) as well as productive moments of discovery (e.g., coordinated tuning, hypothesis-building, explanatory talk). These observations inform the next iteration by identifying which pacing and interaction conditions best support sustained engagement, collaborative interpretation, and thoughtful attention to marine pollution without relying on explicit instructions or didactic explanation.

We are particularly attentive to whether participants move through an implicit engagement arc from novel attraction, to

committed exploration, to critical contemplation, and, in some cases, to expressions of concern or responsibility. These observations will validate current design assumptions and inform future iterations, indicating where the feedback system and narrative inference are supported, ambiguous, or misaligned.

6 Discussion

By staging a designed contrast between slow, open-ended exploration and time-pressured, replayable play, *In Deep Water* engages ongoing debates within HCI and game design regarding attention, care, and engagement. Previous work suggests that temporal framing and performance structures can substantially alter how players interpret meaning and coordinate action, even when interacting with the same underlying system [2, 7, 15].

The demo does not seek to resolve these tensions, but to make them experientially available. In doing so, it positions interactive installations as spaces for collective reflection on how design choices shape not only interaction, but also the values and assumptions embedded in public engagement with environmental research.

7 Conclusion

In Deep Water positions games and interactive simulations as practice platforms for engaging with environmental complexity. By balancing the playful affordances of game technologies with the expertise of game designers and the seriousness of scientific research, the project explores how players might come to understand themselves as situated actors within real-world ecosystems. The proposed CHI 2026 demo aligns with the conference theme *Creating Tomorrow Together* by treating interaction as a shared interpretive process rather than an individual task. Through collaborative exploration, embodied decoding, and comparative engagement, *In Deep Water* demonstrates how games can contribute to imagining and enacting more responsible futures, inviting players not only to learn about environmental harm, but to rehearse ways of noticing, caring, and acting together.

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